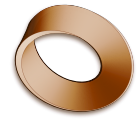


L2 – Plotting Shapes

Unit 3a: Interactive Animations and Games



Design 9B • Mr. Rawson • 13 students

Date	Friday 28 August 2026	Room	PH1
Period	P6, 15:05–15:55	Platform	CodeAI, Game Lab
Lesson	2 of 28, unplugged	Next lesson	Wed 9 Sep, P5
Criterion	Pre-summative, feeds B	Unit map	_Unit Map.md
Homework	L3 <i>Drawing in Game Lab</i> , 9 levels, due Wed 9 Sep. L8 is assessed.		

Learning intentions

By the end of the lesson students can:

- say where the origin is and which way y increases;
- give instructions precise enough for someone else to reproduce a drawing;
- explain why the *order* of instructions changes the result.

Timing

Time	Phase	What happens
before P6	Printing	Print 7 of Version A , 6 of Version B . One device each, roster already synced.
15:05–15:10	Settle	Kit check. AI rule restated: own thinking, AI scores zero.
15:10–15:20	Sign in	Sign in via Google Classroom. Open Unit 3a L2, the Drawing Tool.
15:20–15:45	Paired activity	Sheets face down. Partners back to back. A describes, B builds. Swap at 15:33.
15:45–15:55	Reflect	Page 2 questions, short share out, set homework.

Questions to ask

1. Where is $(0,0)$, and what happens to y as you move down the page?
2. “Put a red circle in the middle.” Give me three drawings that obey that.
3. Two shapes overlap. What else must you tell your partner?
4. How is this the same as writing code?

Common misconceptions

- “ **y goes up.**” Origin is top left, y increases downward. Put the axes up first.
- “**A coordinate marks the middle.**” True of circles. A square is placed by its top left corner.
- “**Order does not matter.**” In Drawing 2A a red circle sits under a blue square.

Differentiation

Support: stay on Drawing 1 and read grid values aloud.

Extend: invent a drawing on page 2, or cap the describer at twenty words.

Name _____ Period _____ Date _____

Activity Guide - Plotting Shapes A

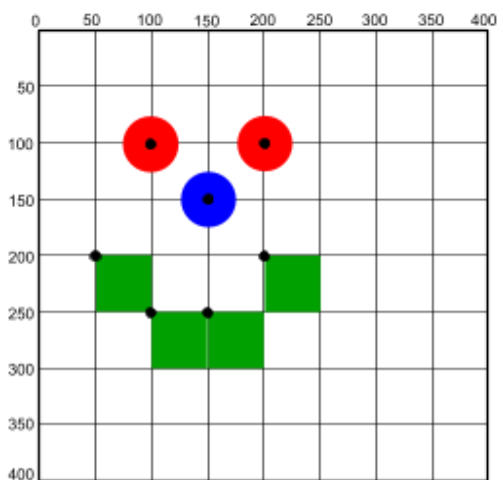


IMPORTANT!! DON'T LET YOUR PARTNER SEE THIS PAPER!

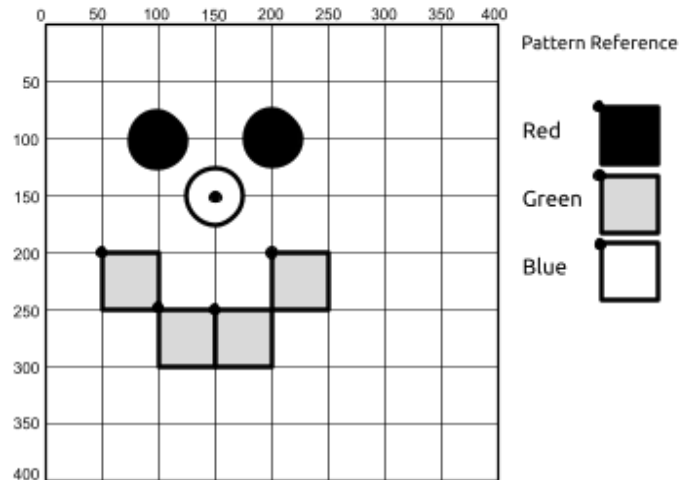
Overview

Your partner should have the Drawing Tool open on a computer where you cannot see it. Alternate turns trying explaining how to draw your image. Afterwards check their work but make sure to keep your drawings hidden.

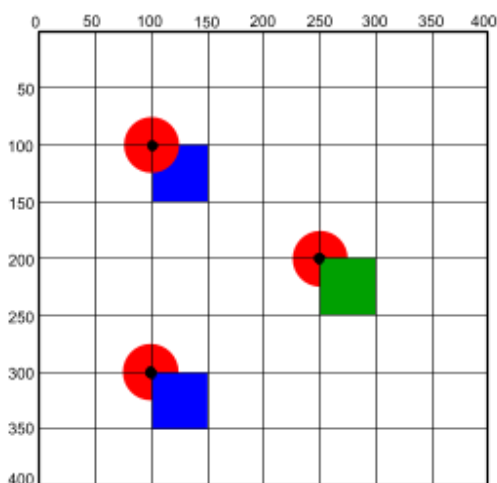
Drawing 1A (Color)



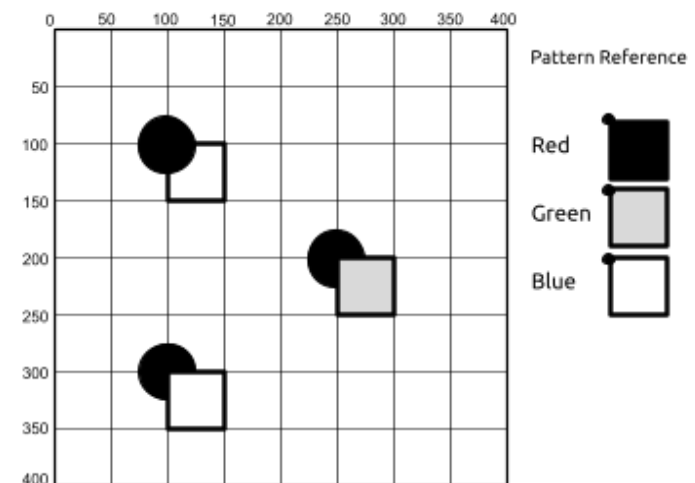
Drawing 1A (Black and White)



Drawing 2A (Color)

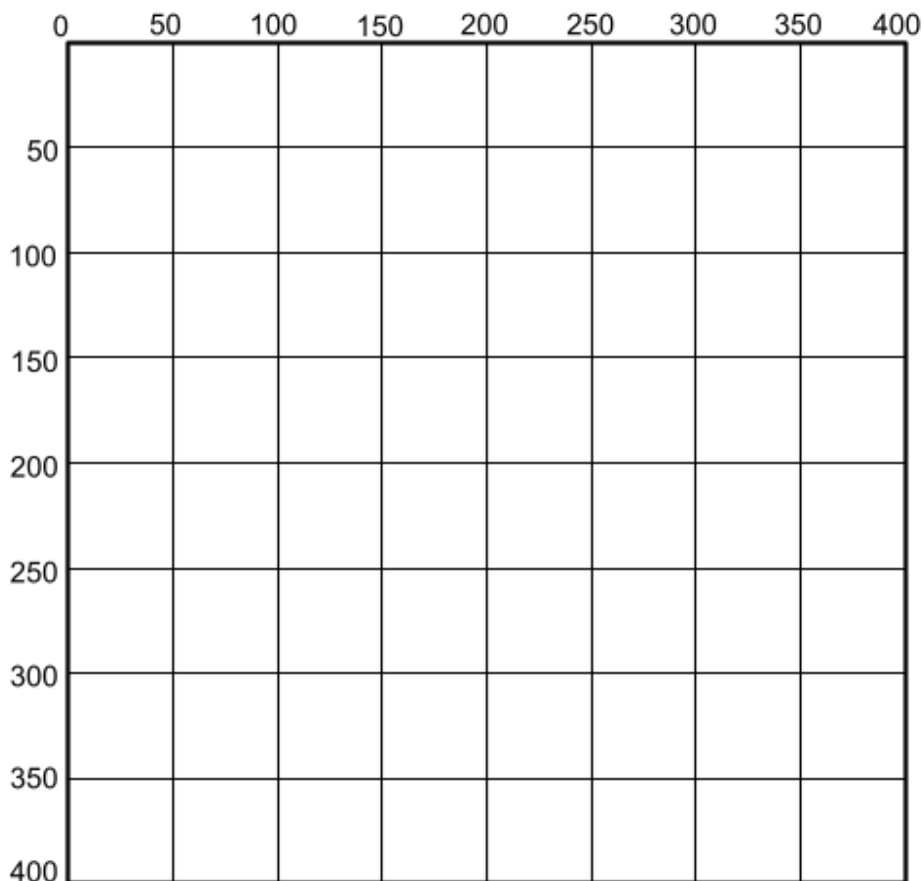


Drawing 2A (Black and White)



Draw Your Own

Use the space below to draw your own image with the shapes. Then see if you can communicate it to your partner to draw using the shape drawing tool in Game Lab. You can also give your drawing to another group to use as a challenge.



Shape Size

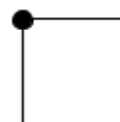
These shapes are the correct size



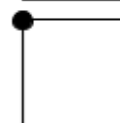
Pattern Reference

If you don't have red, green, or blue to draw with fill in the patterns or colors you'll use instead

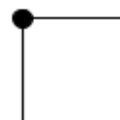
Red



Green



Blue



Reflect

What things were important in communicating about position, color, and order of the shapes in this activity?

What's another way you have seen similar problems solved in the past?

Name _____ Period _____ Date _____

Activity Guide - Plotting Shapes B

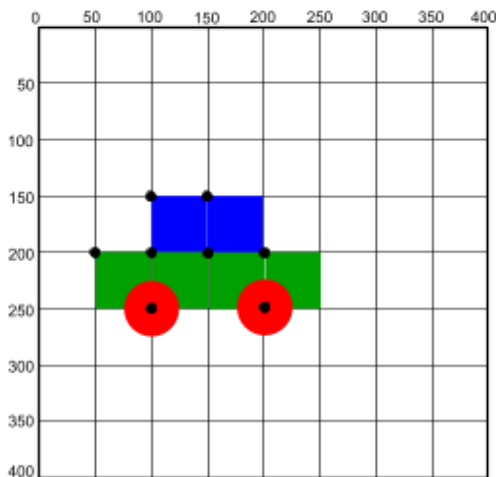


IMPORTANT!! DON'T LET YOUR PARTNER SEE THIS PAPER!

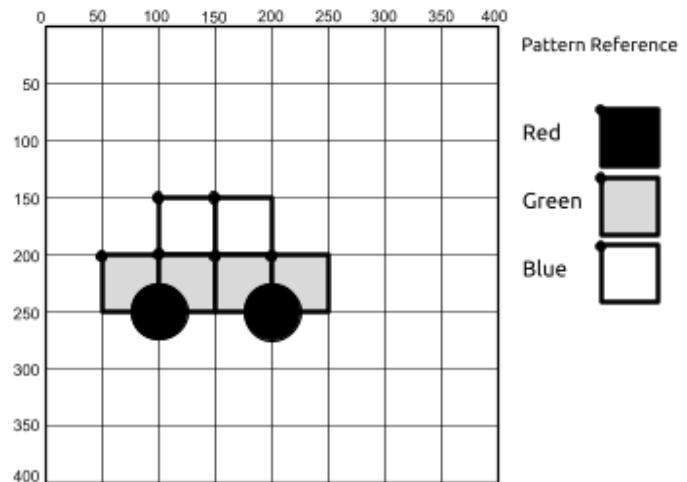
Overview

Your partner should have the Drawing Tool open on a computer where you cannot see it. Alternate turns trying explaining how to draw your image. Afterwards check their work but make sure to keep your drawings hidden.

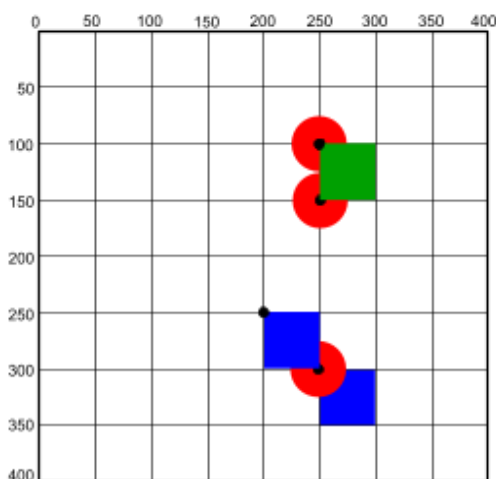
Drawing 1B (Color)



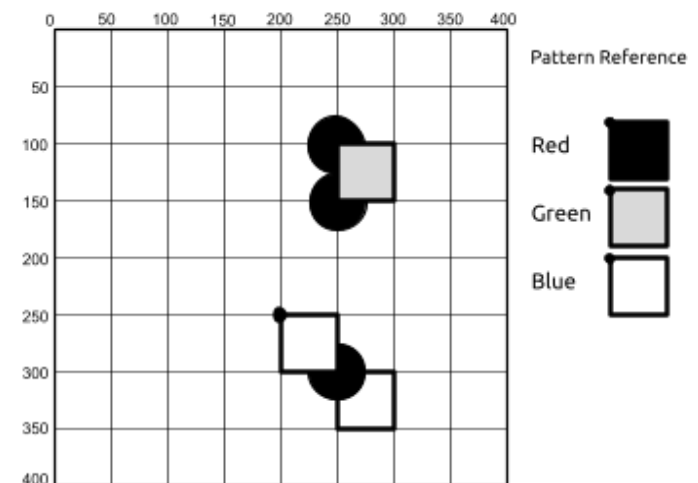
Drawing 1B (Black and White)



Drawing 2B (Color)

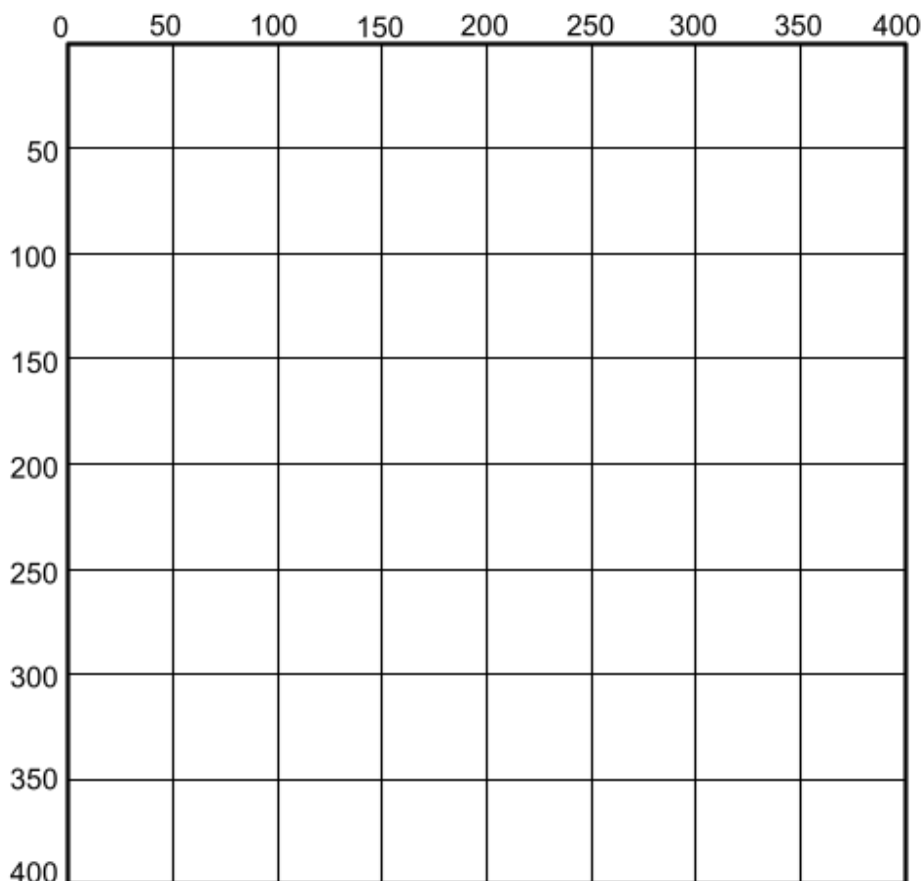


Drawing 2B (Black and White)



Draw Your Own

Use the space below to draw your own image with the shapes. Then see if you can communicate it to your partner to draw using the shape drawing tool in Game Lab. You can also give your drawing to another group to use as a challenge.



Shape Size

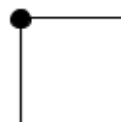
These shapes are the correct size



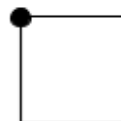
Pattern Reference

If you don't have red, green, or blue to draw with fill in the patterns or colors you'll use instead

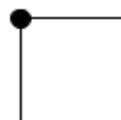
Red



Green



Blue



Reflect

What things were important in communicating about position, color, and order of the shapes in this activity?

What's another way you have seen similar problems solved in the past?
